

INDIAN STATISTICAL INSTITUTE
Bachelor of Statistical Data Science, Third Year
First Semester, 2026-27
Statistical Learning I
Mid-semester Examination Solutions

Combined Solutions for Part-A and Part-B

Part-A Solutions

Problem 1

Question Summary: A student attempts to estimate multiple linear regression coefficients $\beta_0, \beta_1, \beta_2$ of Y on X_1, X_2 using two successive simple linear regressions, where $\sum X_1 = \sum X_2 = \sum X_1 X_2 = 0$.

Solution:

Since $\sum_{i=1}^n X_{1i} = \sum_{i=1}^n X_{2i} = 0$, both X_1 and X_2 have sample means of 0 ($\bar{X}_1 = \bar{X}_2 = 0$). Also, since $\sum X_{1i} X_{2i} = 0$, the predictors are strictly orthogonal.

In the true multiple linear regression $\hat{Y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2$, the OLS estimates under orthogonality are:

$$\begin{aligned}\beta_0 &= \bar{Y} \\ \beta_1 &= \frac{\sum X_{1i} Y_i}{\sum X_{1i}^2} \\ \beta_2 &= \frac{\sum X_{2i} Y_i}{\sum X_{2i}^2}\end{aligned}$$

The student performs two simple regressions. First, regressing Y on X_1 :

$$\begin{aligned}\alpha_1 &= \frac{\sum (X_{1i} - \bar{X}_1)(Y_i - \bar{Y})}{\sum (X_{1i} - \bar{X}_1)^2} = \frac{\sum X_{1i} Y_i}{\sum X_{1i}^2} = \beta_1 \\ \alpha_0 &= \bar{Y} - \alpha_1 \bar{X}_1 = \bar{Y} = \beta_0\end{aligned}$$

The residuals are $Y_i^* = Y_i - \alpha_0 - \alpha_1 X_{1i} = Y_i - \bar{Y} - \beta_1 X_{1i}$. The sample mean is $\bar{Y}^* = 0$.

Next, the student regresses the residuals Y^* on X_2 :

$$\begin{aligned}\gamma_2 &= \frac{\sum X_{2i} Y_i^*}{\sum X_{2i}^2} = \frac{\sum X_{2i} (Y_i - \bar{Y} - \beta_1 X_{1i})}{\sum X_{2i}^2} \\ &= \frac{\sum X_{2i} Y_i - \bar{Y} \sum X_{2i} - \beta_1 \sum X_{1i} X_{2i}}{\sum X_{2i}^2} = \frac{\sum X_{2i} Y_i}{\sum X_{2i}^2} = \beta_2\end{aligned}$$

The intercept is $\gamma_0 = \bar{Y}^* - \gamma_2 \bar{X}_2 = 0 - 0 = 0$.

The student concludes:

$$\begin{aligned}\beta_0^{\text{student}} &= \alpha_0 + \gamma_0 = \bar{Y} = \beta_0 \\ \beta_1^{\text{student}} &= \alpha_1 = \beta_1 \\ \beta_2^{\text{student}} &= \gamma_2 = \beta_2\end{aligned}$$

The student derived the exact correct coefficients. They should receive the full **6 marks**.

Problem 2

Question Summary: Find θ that minimizes $\sum_{i=1}^8 |Y_i - \theta X_i|$.

Solution:

For $X_i \neq 0$, rewrite the absolute deviation as $|X_i| \left| \frac{Y_i}{X_i} - \theta \right|$. For $X_i = 0$ ($X_4 = 0, Y_4 = 1$), the term is $|1| = 1$, independent of θ .

Minimizing the objective is finding the weighted median of ratios $Z_i = \frac{Y_i}{X_i}$ with weights $W_i = |X_i|$ for $i \neq 4$.

X_i	Y_i	$Z_i = Y_i/X_i$	Weight $W_i = X_i $	Cumulative Weight
-1	0	0	1	1
-2	-3	1.5	2	3
-3	-5	$5/3 \approx 1.67$	3	6
4	8	2	4	10
3	7	$7/3 \approx 2.33$	3	13
2	5	2.5	2	15
1	3	3	1	16

The total weight is 16. The weighted median occurs where cumulative weight reaches $16/2 = 8$. It hits 10 at $Z = 2$. Therefore, $\theta = 2$.

Problem 3

Question Summary: Minimize sum of squares given orthogonal design $\mathbf{X}^\top \mathbf{X} = \mathbf{I}_2$ and OLS fit $\hat{\beta}^{\text{LS}} = (1.2, -0.5)^\top$, subject to (a) $|\beta_1| + |\beta_2| \leq 0.5$ and (b) $0.5 \leq \beta_1^2 + \beta_2^2 \leq 1$.

Solution:

(a) Lasso Constraint ($|\beta_1| + |\beta_2| \leq 0.5$)

Because $\mathbf{X}^\top \mathbf{X} = \mathbf{I}_2$, the Lasso estimate for each coefficient is given by the soft-thresholding operator:

$$\hat{\beta}_j^{\text{Lasso}} = \text{sign}(\hat{\beta}_j^{\text{LS}}) \max\left(|\hat{\beta}_j^{\text{LS}}| - \lambda, 0\right)$$

We choose λ to satisfy the constraint. Since the unconstrained OLS sum is $|1.2| + |-0.5| = 1.7 > 0.5$, the constraint is binding, and we must solve:

$$\max(1.2 - \lambda, 0) + \max(|-0.5| - \lambda, 0) = 0.5$$

We test $\lambda > 0.5$ so the second term zeroes out:

$$\max(1.2 - \lambda, 0) + 0 = 0.5 \implies 1.2 - \lambda = 0.5 \implies \lambda = 0.7$$

This $\lambda = 0.7$ satisfies both terms perfectly. Substituting $\lambda = 0.7$ back into the estimators yields:

$$\begin{aligned}\beta_1 &= \text{sign}(1.2) \max(1.2 - 0.7, 0) = 0.5 \\ \beta_2 &= \text{sign}(-0.5) \max(0.5 - 0.7, 0) = 0\end{aligned}$$

Thus, $\beta_1 = 0.5, \beta_2 = 0$.

(b) Ridge Constraint ($0.5 \leq \beta_1^2 + \beta_2^2 \leq 1$)

The Ridge regression estimator is derived from:

$$\hat{\beta}^{\text{Ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{Y}$$

Since $\mathbf{X}^\top \mathbf{X} = \mathbf{I}_2$, this simplifies to a scaled version of the OLS estimate:

$$\hat{\beta}^{\text{Ridge}} = (\mathbf{I} + \lambda \mathbf{I})^{-1} \hat{\beta}^{\text{LS}} = \frac{1}{1 + \lambda} \hat{\beta}^{\text{LS}} = c \hat{\beta}^{\text{LS}}$$

We know that as the estimate moves away from the OLS estimate toward the origin, the squared error increases. Thus, we want to choose c (or λ) to shrink the estimate as little as possible while satisfying the constraint. The squared Euclidean norm of the OLS estimate is $(1.2)^2 + (-0.5)^2 = 1.69$, which is strictly greater than 1. Therefore, the closest boundary of the feasible region is the outer boundary $\beta_1^2 + \beta_2^2 = 1$. Setting the squared norm to 1:

$$\|c \hat{\beta}^{\text{LS}}\|^2 = 1 \implies c^2(1.69) = 1 \implies c = \frac{1}{\sqrt{1.69}} = \frac{1}{1.3} = \frac{10}{13}$$

Therefore, the estimates are:

$$\beta_1 = \frac{10}{13}(1.2) = \frac{12}{13}, \quad \beta_2 = \frac{10}{13}(-0.5) = -\frac{5}{13}$$

Thus, $\beta_1 = 12/13, \beta_2 = -5/13$.

Part-B Solutions

Problem 1

Question Summary: Evaluate missing Y values a_1 and a_2 using Nadaraya-Watson limit bounds $\lim_{h \downarrow 0} \hat{f}_h(2) = 4$ and $\lim_{h \uparrow \infty} \hat{f}_h(2) = 3.6$.

Solution:

For a dataset (X_i, Y_i) , the Nadaraya-Watson kernel estimate is:

$$\hat{f}_h(x) = \frac{\sum_{i=1}^n K_h(x - X_i) Y_i}{\sum_{i=1}^n K_h(x - X_i)}$$

As $h \downarrow 0$, the kernel places weight strictly on the closest point(s). The closest data point X_i to $x = 2$ is 1.8 (distance = 0.2, next closest is 2.3 with distance 0.3). Thus, $\lim_{h \downarrow 0} \hat{f}_h(2)$ simplifies to the Y value at $X = 1.8$, which is a_1 . Given $\lim_{h \downarrow 0} \hat{f}_h(2) = 4$, we have $a_1 = 4$.

As $h \uparrow \infty$, the kernel $K_h(x - X_i) \rightarrow K(0)$ for all i , yielding equal weights. Thus, $\lim_{h \uparrow \infty} \hat{f}_h(x) = \bar{Y}$, the sample mean of Y . We are given $\bar{Y} = 3.6$. Over $n = 10$ observations, the sum of Y must be 36. Summing the given Y values:

$$3.6 + 3.3 + 3.1 + 2.9 + 3.3 + 3.6 + 4 + a_2 + 4.1 + 3.6 = 31.5 + a_2$$

Setting $31.5 + a_2 = 36$ yields $a_2 = 4.5$.

Problem 2

Question Summary: Provide concrete examples for three regression/classification scenarios.

Solution:

(a) Projection Pursuit expected to yield much better than Additive Regression:

Consider a model with a pure interaction effect: $Y = X_1 X_2 + \epsilon$. Additive regression $f(x) = f_1(X_1) + f_2(X_2)$ cannot natively capture this interaction. Projection Pursuit Regression (PPR) models $f(x) = \sum_{m=1}^M g_m(w_m^\top x)$. By applying the algebraic identity $X_1 X_2 = \frac{1}{4}(X_1 + X_2)^2 - \frac{1}{4}(X_1 - X_2)^2$, PPR captures the relationship perfectly using $M = 2$ ridge functions along directions $(1, 1)^\top$ and $(1, -1)^\top$.

(b) Continuous class distributions but Bayes rule is not unique:

Let priors be equal: $\pi_1 = \pi_2 = 1/2$. Let $f_1 \sim \text{Uniform}(0, 2)$ and $f_2 \sim \text{Uniform}(1, 3)$. The Bayes classifier assigns an observation to the class with the maximum posterior probability. For any point in the overlapping region $x \in (1, 2)$, the posterior probabilities are identical: $P(\text{Class 1} | x) = P(\text{Class 2} | x) = 1/2$. Thus, any arbitrary classification rule on $(1, 2)$ yields exactly the same misclassification rate. The Bayes rule is therefore not uniquely defined.

(c) Continuous distributions where no classifier can have error smaller than 1/3:

Let priors be equal: $\pi_1 = \pi_2 = 1/2$. Let the class-conditional distributions be heavily overlapping: $f_1 \sim \text{Uniform}(0, 3)$ and $f_2 \sim \text{Uniform}(1, 4)$.

Let's evaluate the explicit probability of error of the Bayes Classifier, which sets the absolute lower bound for any classifier:

- For $x \in (0, 1)$, only Class 1 occurs ($f_2 = 0$). The classifier picks Class 1. Error = 0.
- For $x \in (1, 3)$, both classes have equal density $f_1(x) = f_2(x) = 1/3$. The classifier can arbitrarily assign to either class (say, Class 1). The error here is the probability that the actual class was Class 2, which is $\int_1^3 \pi_2 f_2(x) dx = \frac{1}{2} \int_1^3 \frac{1}{3} dx = \frac{1}{2} \times \frac{2}{3} = \frac{1}{3}$.
- For $x \in (3, 4)$, only Class 2 occurs ($f_1 = 0$). The classifier picks Class 2. Error = 0.

The total Bayes misclassification risk is $0 + 1/3 + 0 = 1/3$. Because the optimal Bayes Risk is $1/3$, no classifier can achieve a smaller error probability.

Problem 3

Question Summary: Best binary split for a categorical variable $C \in \{C_1, C_2, C_3\}$ for every concave and symmetric impurity function.

Solution:

The total counts are 500 (250 in Class 1 and 250 in Class 2). The proportions p of Class 1 are: C_1 : 1.0, C_3 : 0.5, C_2 : $100/275 \approx 0.364$. Because $1.0 > 0.5 > 0.364$, the only two admissible binary splits are:

- **Split 1** ($\{C_1\}$ vs $\{C_2, C_3\}$): $p_A^{(1)} = 1$ (weight 15%), $p_B^{(1)} = 7/17 \approx 0.41$ (weight 85%).
- **Split 2** ($\{C_1, C_3\}$ vs $\{C_2\}$): $p_A^{(2)} = 2/3$ (weight 45%), $p_B^{(2)} = 4/11 \approx 0.364$ (weight 55%).

Since an impurity function ψ is symmetric about 0.5, we know that $\psi(1) = \psi(0)$ and $\psi(2/3) = \psi(1/3)$. For Split 1, the probabilities are 1 (weight 0.15) and 7/17 (weight 0.85). For Split 2, the probabilities are 2/3 (weight 0.45) and 4/11 (weight 0.55).

We want to show that Split 1 yields a larger decrease in impurity than Split 2 for **any** strictly concave function ψ . Using the symmetry property $\psi(2/3) = \psi(1/3)$ and $\psi(1) = \psi(0)$, we can express the relevant evaluation points of Split 2 directly as convex combinations of the evaluation points corresponding to Split 1 and apply **Jensen's Inequality**:

$$\begin{aligned} 1/3 &= \frac{4}{21}(0) + \frac{17}{21}\left(\frac{7}{17}\right) \implies \psi\left(\frac{2}{3}\right) = \psi\left(\frac{1}{3}\right) > \frac{4}{21}\psi(0) + \frac{17}{21}\psi\left(\frac{7}{17}\right) = \frac{4}{21}\psi(1) + \frac{17}{21}\psi\left(\frac{7}{17}\right) \\ 4/11 &= \frac{9}{77}(0) + \frac{68}{77}\left(\frac{7}{17}\right) \implies \psi\left(\frac{4}{11}\right) > \frac{9}{77}\psi(0) + \frac{68}{77}\psi\left(\frac{7}{17}\right) = \frac{9}{77}\psi(1) + \frac{68}{77}\psi\left(\frac{7}{17}\right) \end{aligned}$$

Substituting these into the Expected Impurity for Split 2 yields:

$$\begin{aligned} \text{Impurity}(\text{Split 2}) &= 0.45 \psi(2/3) + 0.55 \psi(4/11) \\ &> 0.45 \left[\frac{4}{21} \psi(1) + \frac{17}{21} \psi\left(\frac{7}{17}\right) \right] + 0.55 \left[\frac{9}{77} \psi(1) + \frac{68}{77} \psi\left(\frac{7}{17}\right) \right] \\ &= \psi(1) \left[\frac{9}{20} \cdot \frac{4}{21} + \frac{11}{20} \cdot \frac{9}{77} \right] + \psi(7/17) \left[\frac{9}{20} \cdot \frac{17}{21} + \frac{11}{20} \cdot \frac{68}{77} \right] \\ &= \psi(1) \left[\frac{12}{140} + \frac{9}{140} \right] + \psi(7/17) \left[\frac{51}{140} + \frac{68}{140} \right] \\ &= 0.15 \psi(1) + 0.85 \psi(7/17) \\ &= \text{Impurity}(\text{Split 1}) \end{aligned}$$

Jensen's inequality explicitly proves that the expected impurity of Split 2 is strictly greater than the expected impurity of Split 1. Therefore, **Split 1** ($\{C_1\}$ vs $\{C_2, C_3\}$) provides the maximum decrease universally.